Step 1 - Read the JSON file using the spark dataframe reader API

```
dbutils.widgets.text("p_data_source", "")
v_data_source = dbutils.widgets.get("p_data_source")
```

```
dbutils.widgets.text("p_file_date", "2021-03-21")
v_file_date = dbutils.widgets.get("p_file_date")
```

```
%run "../includes/configuration"
```

```
%run "../includes/common_functions"
```

```
from pyspark.sql.types import StructType, StructField, IntegerType, StringType, DateType
```

```
name_schema = StructType(fields=[StructField("forename", StringType(), True),
                                 StructField("surname", StringType(), True)

])
```

```
drivers_schema = StructType(fields=[StructField("driverId", IntegerType(), False),
                                    StructField("driverRef", StringType(), True),
                                    StructField("number", IntegerType(), True),
                                    StructField("code", StringType(), True),
                                    StructField("name", name_schema),
                                    StructField("dob", DateType(), True),
                                    StructField("nationality", StringType(), True),
                                    StructField("url", StringType(), True)
])
```

```
drivers_df = spark.read \
.schema(drivers_schema) \
.json(f"{raw_folder_path}/{v_file_date}/drivers.json")
```

```
display(drivers_df)
```



	driverId	driverRef	number	code	name	dob	na
1	1	hamilton	44	HAM	{"forename":"Lewis","surname":"Hamilton"}	1985-01-07	British
2	2	heidfeld	null	HEI	{"forename":"Nick","surname":"Heidfeld"}	1977-05-10	Germa
3	3	rosberg	6	ROS	{"forename":"Nico","surname":"Rosberg"}	1985-06-27	Germa
4	4	alonso	14	ALO	{"forename":"Fernando","surname":"Alonso"}	1981-07-29	Spanis
5	5	kovalainen	null	KOV	{"forename":"Heikki","surname":"Kovalainen"}	1981-10-19	Finnish
6	6	nakajima	null	NAK	{"forename":"Kazuki","surname":"Nakajima"}	1985-01-11	Japane
7	7	bourdais	null	BOU	{"forename":"Sébastien","surname":"Bourdais"}	1979-02-28	Frenc
8	8	raikkonen	7	RAI	{"forename":"Kimi","surname":"Räikkönen"}	1979-10-17	Finnish
9	9	kubica	88	KUB	{"forename":"Robert","surname":"Kubica"}	1984-12-07	Polish
10	10	glock	null	GLO	{"forename":"Timo","surname":"Glock"}	1982-03-18	Germa
11	11	sato	null	SAT	{"forename":"Takuma","surname":"Sato"}	1977-01-28	Japane
12	12	piquet_jr	null	PIQ	{"forename":"Nelson","surname":"Piquet Jr."}	1985-07-25	Brazili
13	13	massa	19	MAS	{"forename":"Felipe","surname":"Massa"}	1981-04-25	Brazili
14	14	coulthard	null	COU	{"forename":"David","surname":"Coulthard"}	1971-03-27	British

853 rows

Step 2 - Rename columns and add new columns

- driverId renamed to driver_id
- driverRef renamed to driver_ref
- ingestion date added
- name added with concatenation of forename and surname

```
from pyspark.sql.functions import col, concat, current_timestamp, lit
```

```
drivers_with_ingestion_date_df = add_ingestion_date(drivers_df)
```

```
drivers_with_columns_df = drivers_df.withColumnRenamed("driverId", "driver_id") \
                                    .withColumnRenamed("driverRef", "driver_ref") \
                                    .withColumn("ingestion_date", current_timestamp()) \
                                    .withColumn("name", concat(col("name.forename"), lit(" "), col("name.surname"))) \
                                    .withColumn("data_source", lit(v_data_source)) \
                                    .withColumn("file_date", lit(v_file_date))
```

```
display(drivers_with_columns_df)
```

Table

	driver_id	driver_ref	number	code	name	dob	nationality	url
1	1	hamilton	44	HAM	Lewis Hamilton	1985-01-07	British	http://en
2	2	heidfeld	null	HEI	Nick Heidfeld	1977-05-10	German	http://en
3	3	rosberg	6	ROS	Nico Rosberg	1985-06-27	German	http://en
4	4	alonso	14	ALO	Fernando Alonso	1981-07-29	Spanish	http://en
5	5	kovalainen	null	KOV	Heikki Kovalainen	1981-10-19	Finnish	http://en
6	6	nakajima	null	NAK	Kazuki Nakajima	1985-01-11	Japanese	http://en
7	7	bourdais	null	BOU	Sébastien Bourdais	1979-02-28	French	http://en
8	8	raikkonen	7	RAI	Kimi Räikkönen	1979-10-17	Finnish	http://en
9	9	kubica	88	KUB	Robert Kubica	1984-12-07	Polish	http://en

10	10	glock	null	GLO	Timo Glock	1982-03-18	German	http://en
11	11	sato	null	SAT	Takuma Sato	1977-01-28	Japanese	http://en
12	12	piquet_jr	null	PIQ	Nelson Piquet Jr.	1985-07-25	Brazilian	http://en
13	13	massa	19	MAS	Felipe Massa	1981-04-25	Brazilian	http://en
14	14	coulthard	null	COU	David Coulthard	1971-03-27	British	http://en

853 rows

Step 3 - Drop the unwanted columns

- name.forename
- name.surname
- url

```
drivers_final_df = drivers_with_columns_df.drop(col("url"))
```

```
display(drivers_final_df)
```

Table

	driver_id	driver_ref	number	code	name	dob	nationality	inge
1	1	hamilton	44	HAM	Lewis Hamilton	1985-01-07	British	2024-07
2	2	heidfeld	null	HEI	Nick Heidfeld	1977-05-10	German	2024-07
3	3	rosberg	6	ROS	Nico Rosberg	1985-06-27	German	2024-07
4	4	alonso	14	ALO	Fernando Alonso	1981-07-29	Spanish	2024-07
5	5	kovalainen	null	KOV	Heikki Kovalainen	1981-10-19	Finnish	2024-07
6	6	nakajima	null	NAK	Kazuki Nakajima	1985-01-11	Japanese	2024-07
7	7	bourdais	null	BOU	Sébastien Bourdais	1979-02-28	French	2024-07
8	8	raikkonen	7	RAI	Kimi Räikkönen	1979-10-17	Finnish	2024-07
9	9	kubica	88	KUB	Robert Kubica	1984-12-07	Polish	2024-07
10	10	glock	null	GLO	Timo Glock	1982-03-18	German	2024-07
11	11	sato	null	SAT	Takuma Sato	1977-01-28	Japanese	2024-07
12	12	piquet_jr	null	PIQ	Nelson Piquet Jr.	1985-07-25	Brazilian	2024-07
13	13	massa	19	MAS	Felipe Massa	1981-04-25	Brazilian	2024-07
14	14	coulthard	null	COU	David Coulthard	1971-03-27	British	2024-07

853 rows

Step 4 - Write to output to processed container in parquet format

```
# drivers_final_df.write.mode("overwrite").format("parquet").saveAsTable("f1_processed.drivers")
```

```
drivers_final_df.write.mode("overwrite").format("delta").saveAsTable("f1_processed.drivers")
```

```
# display(spark.read.parquet(f"{processed_folder_path}/drivers"))
```

Table

	driver_id	driver_ref	number	code	name	dob	nationality	inge
1	1	hamilton	44	HAM	Lewis Hamilton	1985-01-07	British	2024-07
2	2	heidfeld	null	HEI	Nick Heidfeld	1977-05-10	German	2024-07
3	3	rosberg	6	ROS	Nico Rosberg	1985-06-27	German	2024-07
4	4	alonso	14	ALO	Fernando Alonso	1981-07-29	Spanish	2024-07
5	5	kovalainen	null	KOV	Heikki Kovalainen	1981-10-19	Finnish	2024-07
6	6	nakajima	null	NAK	Kazuki Nakajima	1985-01-11	Japanese	2024-07
7	7	bourdais	null	BOU	Sébastien Bourdais	1979-02-28	French	2024-07
8	8	raikkonen	7	RAI	Kimi Räikkönen	1979-10-17	Finnish	2024-07
9	9	kubica	88	KUB	Robert Kubica	1984-12-07	Polish	2024-07
10	10	glock	null	GLO	Timo Glock	1982-03-18	German	2024-07
11	11	sato	null	SAT	Takuma Sato	1977-01-28	Japanese	2024-07
12	12	piquet_jr	null	PIQ	Nelson Piquet Jr.	1985-07-25	Brazilian	2024-07
13	13	massa	19	MAS	Felipe Massa	1981-04-25	Brazilian	2024-07
14	14	coulthard	null	COU	David Coulthard	1971-03-27	British	2024-07

853 rows

```
%fs
ls /mnt/formula1dl1204/processed/drivers
```

Table

	path	name
1	dbfs:/mnt/formula1dl1204/processed/drivers/_delta_log/	_delta_log/
2	dbfs:/mnt/formula1dl1204/processed/drivers/part-00000-f9b542a1-1a36-4b0b-b616-45e82dae5518-c000.snappy.parqu...	part-00000-f9b542a1-1a36-4b0b-b616-45

2 rows

```
%sql
select * from f1_processed.drivers;
```

Table

	driver_id	driver_ref	number	code	name	dob	nationality	inge
1	1	hamilton	44	HAM	Lewis Hamilton	1985-01-07	British	2024-07
2	2	heidfeld	null	HEI	Nick Heidfeld	1977-05-10	German	2024-07
3	3	rosberg	6	ROS	Nico Rosberg	1985-06-27	German	2024-07
4	4	alonso	14	ALO	Fernando Alonso	1981-07-29	Spanish	2024-07
5	5	kovalainen	null	KOV	Heikki Kovalainen	1981-10-19	Finnish	2024-07
6	6	nakajima	null	NAK	Kazuki Nakajima	1985-01-11	Japanese	2024-07
7	7	bourdais	null	BOU	Sébastien Bourdais	1979-02-28	French	2024-07
8	8	raikkonen	7	RAI	Kimi Räikkönen	1979-10-17	Finnish	2024-07
9	9	kubica	88	KUB	Robert Kubica	1984-12-07	Polish	2024-07
10	10	glock	null	GLO	Timo Glock	1982-03-18	German	2024-07
11	11	sato	null	SAT	Takuma Sato	1977-01-28	Japanese	2024-07
12	12	piquet_jr	null	PIQ	Nelson Piquet Jr.	1985-07-25	Brazilian	2024-07
13	13	massa	19	MAS	Felipe Massa	1981-04-25	Brazilian	2024-07

14	14	coulthard	null	COO	David Coulthard	1971-03-27	British	2024-07-

853 rows

Success